

SURBHIT KUMAR

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SUMMARY

Software Engineer specializing in Python and TypeScript with expertise in LLM/agent systems, RAG, and model optimization. Delivered production-grade AI/ML services using FastAPI, PyTorch, and TensorFlow at Paradigm Study, and built scalable React/Next.js dashboards for real-time interaction. Designed flexible data pipelines and vector-search services that cut storage costs by 10% at Coupang and reduced return costs by 30% at Flipkart. Seeking to leverage AI engineering and PMP-backed leadership to accelerate product innovation and drive measurable business impact.

PROFESSIONAL EXPERIENCE

Paradigm Study Inc.

Feb 2025 - Present

Software Engineer, Web-ML Integration / AI Agent Development

Cupertino, CA, USA

- Developed FastAPI services that orchestrated AI workflows using the OpenAI Agent SDK, containerized the backend with Docker, and delivered the APIs to a Next.js dashboard, enabling the team to monitor and control AI agents in real time
- Designed and implemented multi-agent pipelines with LangChain, including agent-as-tool and swarm configurations for MCP-style tool-calling, which powered automated textbook creation, question generation, and skill-graph construction, reducing content development time
- Integrated TypeScript/React front-end components that query the FastAPI endpoints and stream responses in real time, reducing user interaction latency and enabling AI-driven features

Coupang

Dec 2018 - Dec 2022

Principal / Senior Manager, Process Innovation

Seoul, South Korea

- Designed and implemented data pipelines to monitor inbound/outbound operations, driving predictive analytics and a 10% reduction in storage costs.
- Led cross-functional teams (up to 15 members) to streamline fulfillment processes with an emphasis on data-driven efficiency.
- Pioneered a culture of continuous improvement, integrating advanced warehouse management tools and analytics solutions.

Flipkart India Pvt Ltd

Mar 2011 - Dec 2018

Senior Program Manager

Bangalore, India

- Drove data-driven improvements in payment operations and supply chain workflows, reducing the Cost of Returns by 30%.
- Collaborated with tech teams to create automation tools and end-to-end supply chain software, resulting in significant cost savings.
- Scaled logistics capacity 5x during large-scale festive sales through strategic technology integrations.

EDUCATION

Rochester Institute of Technology, Rochester, NY, USA

Jan 2023 - May 2025

Master of Science, Computer Science

- Achievements:** Capstone: BODHI – Blended Optimization and Distillation for High-Impact Intelligence, Advanced Graduate Certificate for Artificial Intelligence

Uttar Pradesh Technical University, Lucknow, India

Aug 2004 - Jul 2008

Bachelor of Technology, Computer Science & Engineering

PROJECTS

BODHI: Blended Optimization and Distillation for High-Impact Intelligence

Nov 2024 - May 2025

- Engineered a novel LLM distillation framework optimizing teacher-student models for improved inference efficiency and compute utilization.
- Achieved ~18% improvement in perplexity over untrained student models and strong relative performance compared to teacher models.
- Stack: PyTorch · TensorFlow · FastAPI Docker Flutter(dart).

Multilingual Educational Videos Using LLMS

Apr 2024 - May 2025

- Created a web application to produce LLM-driven educational videos in 5+ languages.
- Python Flutter(dart) · REST APIs · TTS.

AI-Powered E-Commerce Chatbot

Sep 2024 - Dec 2024

- Developed a Python-based Retrieval-Augmented Generation (RAG) chatbot with end-to-end data ingestion & query pipeline.
- Stack: FastAPI · LangChain · OpenAI embeddings · ChromaDB · Flutter(dart).

Stock Price Prediction Using Sentiment Analysis

Feb 2024 - Dec 2024

- Designed a pandas data pipeline to scrape, cleanse, and feed text data for real-time sentiment analysis.
- TensorFlow LSTM · VADER · Python

CERTIFICATIONS

- Certified PMP®: Project Management Institute** | Feb 2015 – Feb 2027
- Machine Learning Specialization: Stanford University** | Aug 2023
- Java Programming: Solving Problems with Software (with Honors): Duke University** | Nov 2017
- Agile Development Specialization: University of Virginia** | Sep 2017

SKILLS

- Programming Languages:** Python, Java, TypeScript, SQL
- Frameworks & Libraries:** FastAPI, TensorFlow, PyTorch, LangChain, React, Next.js, Flutter, NLTK, etc.
- Tools & Technologies:** Git, Docker, AWS, Kubernetes, OpenAI SDK
- Databases:** PostgreSQL, Chroma DB, MongoDB, Neo4j
- Methodologies:** Agile/Scrum, Waterfall
- Other:** Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Machine Learning (ML), LLM training & knowledge distillation, Vector Search